

WASILLA, AK, USA

WMO: 702325

Lat: 61.572N

Lon: 149.541W

Elev: 108

StdP: 100.04

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 26443

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-22.6	-32.7	0.2	-21.1	-31.1	0.2	-18.5	12.1	-4.8	10.5	-3.4	0.6	250	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	23.9	15.6	22.3	14.8	20.2	13.9	16.6	22.9	15.5	20.8	14.5	19.0	2.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.8	10.0	18.8	12.8	9.3	17.2	12.3	9.0	16.7	46.8	22.9	43.5	20.9	40.6	18.8	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.0	6.8	-27.5	26.2	4.7	2.8	-30.9	28.2	-33.7	29.8	-36.3	31.4	-39.8	33.4		
			-27.9	18.1	4.3	1.8	-30.9	19.3	-33.4	20.4	-35.8	21.4	-38.9	22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-8.0	-5.4	-2.8	3.5	8.9	13.1	14.8	13.6	9.1	2.8	-4.6	-7.4
	DBStd	9.71	8.06	6.99	5.65	4.00	3.05	2.60	2.39	2.54	2.92	4.64	6.54	7.13
	HDD10.0	2896	557	432	397	197	56	3	0	2	48	225	438	540
	HDD18.3	5541	816	666	655	446	292	160	114	148	276	482	688	798
	CDD10.0	403	0	0	0	1	22	94	148	113	22	2	0	0
	CDD18.3	7	0	0	0	0	0	1	4	2	0	0	0	0
	CDH23.3	89	0	0	0	0	2	22	44	21	0	0	0	0
	CDH26.7	12	0	0	0	0	0	3	7	2	0	0	0	0
Wind	WSAvg	2.2	2.3	2.4	3.1	2.4	2.1	1.7	1.5	1.6	1.8	2.5	2.5	2.5
Precipitation	PrecAvg	565	34	32	22	13	25	34	53	87	107	59	39	41
	PrecMax	784	71	70	47	40	49	71	101	212	318	125	104	133
	PrecMin	421	5	4	8	1	7	9	13	33	44	17	1	5
	PrecStd	106	20	20	12	10	10	17	23	54	64	32	27	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	7.7	11.1	17.4	22.7	26.3	27.4	26.0	18.7	14.0	8.8	7.4
		MCWB	3.8	3.3	5.1	9.2	12.6	16.9	17.1	16.9	12.2	8.4	5.0	3.4
	2%	DB	6.1	6.2	7.7	13.7	20.0	22.9	24.0	22.8	17.0	11.3	7.4	6.0
		MCWB	2.5	2.3	3.2	6.6	11.2	14.8	16.3	15.7	11.2	6.9	3.8	2.6
	5%	DB	4.0	4.1	6.2	12.0	17.6	21.2	22.2	21.0	15.4	10.0	6.0	3.9
		MCWB	1.0	1.0	1.9	5.3	9.9	14.0	15.4	14.5	10.4	6.0	2.7	1.0
	10%	DB	2.6	2.9	4.1	9.9	16.0	18.9	20.0	18.5	13.8	8.7	3.8	2.5
		MCWB	-0.3	-0.2	0.5	4.3	8.8	12.8	14.6	13.4	9.7	5.3	1.2	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	4.1	5.5	9.6	13.5	17.7	19.0	17.8	13.1	9.1	5.2	4.0
		MCDB	7.4	6.9	9.8	16.9	21.7	24.7	26.4	24.1	17.1	13.4	8.0	6.4
	2%	WB	3.0	2.9	3.9	7.2	11.6	15.6	16.8	16.3	12.1	7.6	4.0	2.8
		MCDB	5.9	6.0	7.3	12.7	18.4	22.0	23.1	22.0	15.3	10.3	6.6	5.7
	5%	WB	1.5	1.7	2.3	5.6	10.4	14.2	15.6	15.0	11.3	6.6	3.0	1.4
		MCDB	4.2	4.5	5.8	10.7	16.7	19.8	21.0	19.6	14.2	9.2	5.5	3.8
	10%	WB	0.0	0.2	1.0	4.5	9.4	13.0	14.8	14.0	10.5	5.5	1.7	-0.2
		MCDB	2.5	3.0	4.6	9.2	15.0	18.3	19.2	17.7	13.2	8.2	3.7	2.2
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.8	8.2	9.3	10.3	9.2	8.0	8.0	7.7	6.5	6.5	6.8
		MCDBR	6.0	6.2	8.6	12.8	14.5	13.3	12.5	11.7	9.7	6.6	5.4	6.4
		MCWBR	4.2	4.6	5.9	7.6	7.7	7.0	6.5	6.2	5.7	4.0	3.7	4.8
	5% WB	MCDBR	5.7	6.0	8.4	11.6	13.0	12.1	11.0	10.2	8.4	6.0	5.0	6.3
		MCWBR	4.1	4.5	5.9	7.1	7.0	6.6	6.1	5.8	5.3	3.9	3.6	5.0
Clear-Sky Solar Irradiance	taub		0.226	0.278	0.298	0.336	0.358	0.358	0.364	0.348	0.310	0.290	0.237	0.377
	taud		2.115	2.248	2.286	2.268	2.300	2.386	2.404	2.462	2.551	2.436	2.098	2.679
	Ebn at Noon		499	701	821	853	861	867	852	838	817	693	458	301
	Edh at Noon		35	63	86	108	114	107	102	89	68	53	35	24
All-Sky Solar Radiation	RadAvg		0.28	0.98	2.61	4.37	5.20	5.33	4.76	3.78	2.45	1.13	0.41	0.14
	RadStd		0.04	0.11	0.24	0.34	0.41	0.34	0.36	0.38	0.31	0.12	0.06	0.02

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	+1.98	+1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page